CHEAT SHEET

What makes a number safe to pay for?

Illustrative case: the company is real, the situation and the data are synthetic, built for this class. Uber, Hyderabad, the driver app. From 31 August 2026 an enrolled driver online through a whole peak window is guaranteed Rs 300 per peak online hour, with a top-up for the gap. Six weeks each side: peak online hours 118,000 to 146,300 (up 24 percent), trips 224,200 to 230,900 (up 3 percent), trips per peak online hour 1.90 to 1.58, offers accepted 0.86 to 0.73, median wait 4.8 to 4.6 minutes, top-up Rs 0 to Rs 1.9 crore. Data: `drivers`, `driver_weeks`, `zone_weeks`. Ask, due Friday: is it working, what should it pay for, is anyone gaming it? Caveats: monsoon's end, 140 new drivers, online means the app is open, three thin outer zones, drivers on two apps.

C2_W04_D01_safe_to_pay_STUDENT
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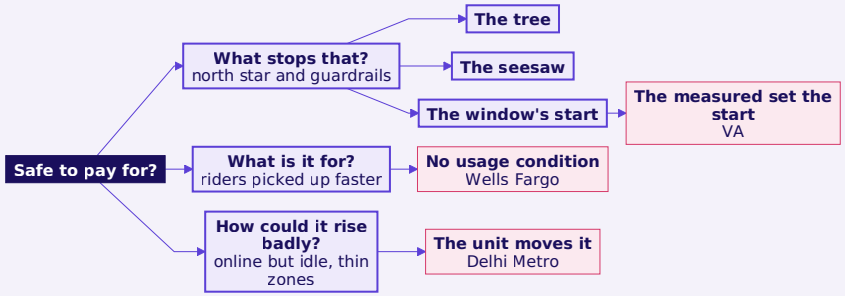
5 panels, 1 diagram

1 What makes a number safe to pay for?



Crux. Pay for completed peak trips per enrolled driver online at peak, guarded by acceptance, the rider's wait and the top-up per trip; hours rose 24 percent while trips per hour fell from 1.90 to 1.58.

2 What did six drivers do, counted by hand?

Driver, zone	Hours, week of 10 Aug then 21 Sep	Trips, the same two weeks
A, Jubilee Hills	10 then 11	21 then 22
B, Banjara Hills	9 then 10	17 then 18
C, Gachibowli	11 then 12	22 then 23
D, Madhapur	10 then 11	19 then 20
E, Shamshabad	10 then 16	18 then 3
F, Rajendranagar	10 then 14	17 then 2
All six	60 then 74, up 23 percent	114 then 88, down 23 percent

Crux. 114 / 60 = 1.90 and 88 / 74 = 1.19: add the trips, add the hours, divide once. Two of six sit in the idle tail against 15 percent of the city, so the city falls less, to 1.58.

3 What are the seven lines of a complete definition?

Line	In plain words	The redesign
Numerator	What is counted	Peak trips completed, one per paying rider
Denominator	Divided by, count printed	Enrolled drivers online one peak hour or more
Window	The stretch of time	The calendar week
Exclusions	Left out, with why	Cancelled trips, test accounts, off-window trips
Does not reward	Behaviour paid nothing	Hours online, declines, a shrinking base
Guardrails	Must not get worse	Accepted share over a floor; median wait; top-up per trip
Gaming note	What each guardrail catches	Declining; lost supply; a runaway top-up

Crux. The first four lines make a number computable; the last three make it hard to game. North star 35.76 to 35.89 trips per online driver-week.

4 What is the proxy test?

Could a team hit this number while the thing it stands for got worse?

Proposed number	How it rises badly	The guardrail
Peak online hours	Online in empty zones	Hours with no trip: 0.22 to 0.38
Completed trips per driver	Short trips only	Offers accepted, over a floor
Enrolled drivers	Sign up, never drive	Share online at peak

Crux. A guardrail is chosen by the gaming it catches; another growth count catches nothing.

5 What did hours cost, and what would the redesign save?

Rs 1.159 crore, 61 percent of the top-up, went to driver-weeks that accepted under half their offers: 180 idle drivers, hours 2.1 times, trips down 12 percent. A floor between 0.46 and 0.78 stops it.

Crux. Saving Rs 58 lakh to Rs 92.7 lakh if a fifth to a half is still paid: an assumption with a range, settled by a test.

THE WORDS ON THIS SHEET

Proxy A number that stands in for the thing wanted

Proxy test Could a team hit this number while the thing it stands for got worse?

North star The one number a plan exists to move

Guardrail A number that must not get worse while the north star rises

Leading input A number that moves before the north star and can be acted on this week

Lagging number A number that confirms the north star after the fact